Study-Wise

CS 3354 Project Milestone P2: Analysis and High-Level Design

Field	Entry
Project title	Study-Wise
Team	Team 02, section .005
Document version	1.0 (baseline for P2; grows from P1 version 0.6)
Date	8 November 2026
Repository URL	https://github.com/shivbatra999/Study-Wise
Git tag for this milestone	p2 (points at the commit that contains this exact PDF)
Submitted on Canvas by	Lokesh Reddy Penugonda

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0 Revision History, Contribution Record and Changes Since P1

Version	Date	Author	Summary of change
0.6	18 Oct 2026	Lokesh Reddy Penugonda	P1 baseline: completed sections 0, 1 and 6 (last P1 row)
1.0	8 Nov 2026	Team	P2 baseline: analysis (sections 4 and 5) and high-level design (sections 6 to 9) added; changes CH-01 to CH-22

0.1 Individual contribution record (P2)

Team member	Sections and artifacts this person produced for P2	Evidence in the repository (commit hashes, PR numbers)	Reviews given (PR numbers)
Lokesh Reddy Penugonda	Benchmark set (`docs/benchmark`); owns sections 1 and 3 (including FR-16 to FR-18), section 4 (use-case analysis and self-review log) and section 9 (traceability); context and use-case figures (3.4)	PR #30 (bb30483), PR #31 (e3c874f),	PR #32 (review comment on spike S-01)
Nandhan Alahari	Spike S-01: wrote and ran the harness on an 8 GB laptop with the 3B and 8B models, committed results and README (PR #32); owns section 8, 6.1 quality-attribute scenarios, and ADR-02, ADR-04 and ADR-06 (first drafted by Lokesh Reddy Penugonda with Claude (10.5); reviewed, revised and committed by this member)	PR #32 (bf4e95e), PR #41 (5a81270), PR #42 (18bf23d)	PR #30, PR #31 (Lesson 13 checklist review of section 4)

Team member	Sections and artifacts this person produced for P2	Evidence in the repository (commit hashes, PR numbers)	Reviews given (PR numbers)
Lavanya Bansal	Hardware and study survey (n = 10), interviews P04–P05 and the elicitation summary; wrote 2.2, 2.3 and section 11 from the results (PRs #36 to #38); owns 0.2 changes since P1, section 2, 10.1 to 10.3 and section 11	PR #36 (b3535c3), PR #37 (22e588a), PR #38 (8a5f0de, 3cfda02),	
Pranab Neupane	Interviews P01–P03 and P06; owns 5.1 domain model and its diagram source, 6.5 analysis-to-architecture mapping, 6.6 design principles and ADR-03	PR #38 (co-author of 8a5f0de, 3cfda02), PR #39 (849dedb), PR #40 (d3899c5), PR #44 (6f72d5e)	
Malik Coleman	Owns 5.2.2 (UC-03 scenario table and sequence diagram source), 5.3 consistency cross-checks K1 to K6, ADR-05 and the 7.2 trade-off summary.		
Shivansh Batra	Owns the cover page, 0.1 record, 5.2.1 (UC-02 scenario table and sequence diagram source), 6.2 to 6.4 (style, component and deployment views and their diagram sources), ADR-01, 10.4 and 10.5, section 12		

0.2 Changes since the P1 baseline

Change ID	What changed (IDs affected)	Why (P1 feedback, new finding, spike result)	Knock-on effect	Agreed on
CH-01	SH1 (2.2), user characteristics (2.3), A1 (2.5) and Q1, Q2, Q4, Q6 to Q8 (11) rewritten from the hardware and study survey (n = 10) and student interviews (n = 5), 6–8 Nov, in `/docs/elicitation`	P1 feedback (C1): the student need was still an assumption from the team's own experience	A1 holds narrowly; the privacy need is not confirmed by students (CON-01 now rests on SH2); FR-17, FR-18 and Q10 follow	8 Nov
CH-02	All guidance boxes and both marking pages removed	P1 feedback (C4)	None	8 Nov
CH-03	Repository URL corrected in 10.4 (P1 had a malformed `%20(` ending)	P1 feedback (C4)	None	7 Nov
CH-04	Team rule: every section merged to main by Sunday 6 PM; PDF committed to `/docs/p2` and tagged p2 before the deadline	P1 feedback (C4): the use-case and traceability file was merged after the P1 deadline	10.4	7 Nov
CH-05	Feasibility spike run as S-01 and moved from `/spike-01` to `/spikes/S-01`; QR-01 verification now cites S-01	P1 feedback: run the spike first so QR-01 rests on measurement; P2 requires the `/spikes` folder	QR-01 confirmed with the 3B model on an 8 GB laptop (section 8); 2.6 and A1 updated; ADR-04	7 Nov
CH-06	FR-16 added: reject a file that cannot be opened (SHOULD)	Analysis finding: UC-01 A3 had no requirement (4.3)	9.1	7 Nov
CH-07	UC-02 and UC-03: P1 pre-conditions "course has materials" and "Study-Wise is set up" became alternate flows; the expanded use cases moved from 3.4 to section 4; old 3.5 matrix replaced by 9.1	Lesson 13 checklist question 6; template structure	4.2, 4.3, 9.1	7 Nov
CH-08	Branch rule `<initials>-<topic>` kept; P1 branches that did not follow it are left as they are, and all P2 branches follow it	Inconsistency between P1 section 4.4 and the P1 contribution record	10.4	7 Nov

Change ID	What changed (IDs affected)	Why (P1 feedback, new finding, spike result)	Knock-on effect	Agreed on
CH-09	FR-15 given a response measure as QAS-04: message within 5 s, no materials lost	Needed a number to make availability a design driver (6.1)	ADR-02	7 Nov
CH-10	Domain model: QuizQuestion gains *student answer*; 4.1 lists UC-03 step (8) as non-trivial	Cross-checks K1 and K4 failed (5.3)	5.1, 5.2.2	7 Nov
CH-11	P1 plan dates that slipped: interviews and survey (planned 26 Oct, started 6 Nov); spike (planned 30 Oct, run 6 Nov); Q3 to the instructor (planned 31 Oct, sent 6 Nov); setup trials (planned 5 Nov, moved to P3)	Work started late; recorded rather than backdated	Risks R1, R3, R4 and questions Q1 to Q5 (10.3, 11)	7 Nov
CH-12	Risk R5 named “Terence Coleman”; corrected to Malik Coleman. Cover version corrected to 1.0	P1 inconsistencies	None	7 Nov
CH-13	Open questions Q7, Q8, Q9 added	Raised by analysis (UC-03, UC-01) and by ADR-04	11	7 Nov
CH-14	ADR-04: default answer model changed from 8B-class to 3B-class (llama3.2:3b); 8B kept only as an option for laptops with at least 16 GB, pending a second S-01 run	S-01: the 8B model did not run on an 8 GB laptop (486 s to load, then the laptop froze); the 3B model answered 20 of 20 within 20 s	QR-03 now at risk (R2); Q9 updated	7 Nov
CH-15	Q3 closed: the instructor confirmed that processing course slides with a local tool for personal study is allowed, and uploading them to a hosted service is not. A4 confirmed; SH2 basis updated; R3 closed	Instructor’s reply to Q3	CON-01 now also rests on the course-materials clause	7 Nov
CH-16	Q6 (marking short answers) returned to the team by the instructor as our own design decision; kept open with options, a decision criterion and a date (section 11)	Instructor’s reply to Q6	FR-11 stays COULD and unmarked until decided	7 Nov
CH-17	ADR-06 added: off-topic questions are refused by a similarity gate followed by a yes/no relevance check, not by a threshold alone (FR-09)	S-01: one off-topic question scored higher than a real benchmark question, so no single threshold separates them	5.2.1 row 4b, 9.1	7 Nov
CH-18	Risk R9 added: some lecture PDFs extract with broken ligatures	S-01 finding on the Chapter 8 deck	FR-03, QR-03; 10.3	7 Nov
CH-19	Q1 answered: 5 of 5 interviewees want a general-knowledge answer clearly labelled as not from their materials. FR-07 and FR-09 kept as materials-only for this baseline; the change is a team decision due 13 Nov (section 11)	Interviews (CH-01)	FR-07, FR-09, QR-03, ADR-06, SH3	8 Nov
CH-20	FR-17 added: keep each submitted quiz and its score for later review (SHOULD); UC-03 postcondition and 6.5 updated	Q7 answered: 3 of 3 interviewees asked want them kept	9.1	8 Nov
CH-21	FR-18 added: replace a material when a file with the same name is imported again, after the student confirms (SHOULD); UC-01 A4 updated	Q8 answered: 3 of 3 interviewees asked said replace, because updated slides often keep the same name	9.1	8 Nov
CH-22	Q10 added (accessibility)	Survey: 2 of 10 respondents use assistive features	2.3, 11	8 Nov

1 to 3 Introduction, Overall Description and Requirements (carried forward)

1 Introduction

1.1 Purpose of this document and of the system

This document is the version 1.0 (P2) baseline of the Study-Wise software requirements specification, extended with use-case analysis and high-level design. It is written for three readers: the CS 3354 instructor and teaching assistants who assess it, the team members who will design, build and test against it, and any student tester who needs to know what the system promises.

Study-Wise addresses a problem students have when they study for exams: their course material, which consists of lecture slides, PDF notes and handouts, is spread across many files, and the AI tools that could help them review it require uploading those files to a third-party cloud service, usually behind a subscription. That upload exposes instructors' material and the student's own questions to an outside company, and many students either cannot pay or do not want to. Study-Wise lets a student ask questions about their own course materials entirely on their own laptop, with every answer pointing back to the slide or page it came from.

1.2 Scope

In scope — we will build this	Out of scope — we will not, and why
Importing text-based PDF and PowerPoint (.pptx) files into courses the student creates	Scanned or handwritten notes that need OCR — accuracy on scans is poor and OCR would consume roughly a third of the schedule
Answering a student's typed question from the materials of one selected course, with a citation to file and page or slide	Cloud sync or access from multiple devices — any server would break the privacy constraint CON-01
Generating multiple-choice practice quizzes from selected materials, scoring them, and showing the source of each answer	Sharing materials or quizzes between students — redistributes instructors' content (stakeholder SH2) and needs user accounts
Keeping courses and imported materials on the device between sessions; removing materials	Integration with UTD eLearning / Canvas — needs institutional API access and credentials the team does not control

Why the exclusions were excluded: each exclusion either conflicts with a constraint or costs more schedule than its value to the core question we are testing. Any of them can be reconsidered through change control once the MUST requirements are met.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
Course	To students, a course is a UTD class they are registered in. In Study-Wise, a course is a named collection of materials the student creates. This document always means the second; it is not linked to registration.
Quiz	In class, a quiz is graded work. In this document, a quiz is an ungraded set of practice questions generated by Study-Wise. Study-Wise never produces or submits graded work (see SH3).
Grounded answer	(a) An answer built only from the student's imported materials; (b) an answer that may also use the model's general knowledge. This document uses (a); the choice is unconfirmed — see Q1.
Material	A PDF or PPTX file the student has imported into a course.
Citation	The file name plus the page number (PDF) or slide number (PPTX) of a passage used in an answer or quiz question.
LLM	Large Language Model — a model that generates text from a prompt.
Local LLM runtime	Software already on the student's computer that runs an LLM and answers requests from other programs on the same machine (Ollama; see A2). It is external to Study-Wise.

Term or abbreviation	Meaning as used in this project
RAG	Retrieval-Augmented Generation — answering by first finding relevant passages in the student's materials and giving them to the LLM with the question.
Benchmark set	20 questions with known answers, written by the team from two real lecture decks and stored in /docs/benchmark; used to verify QR-01 and QR-03 and as the input to spike S-01.
Reference laptop	A team-owned laptop with 16 GB RAM, no discrete GPU, Windows 11 or macOS 14 or later, used for every performance measurement. S-01 records the RAM and operating system of each laptop it runs on.
MoSCoW	Priority scale: MUST, SHOULD, COULD, WON'T (for this release).
PDF / PPTX / OCR	Portable Document Format / PowerPoint Open XML presentation / Optical Character Recognition.

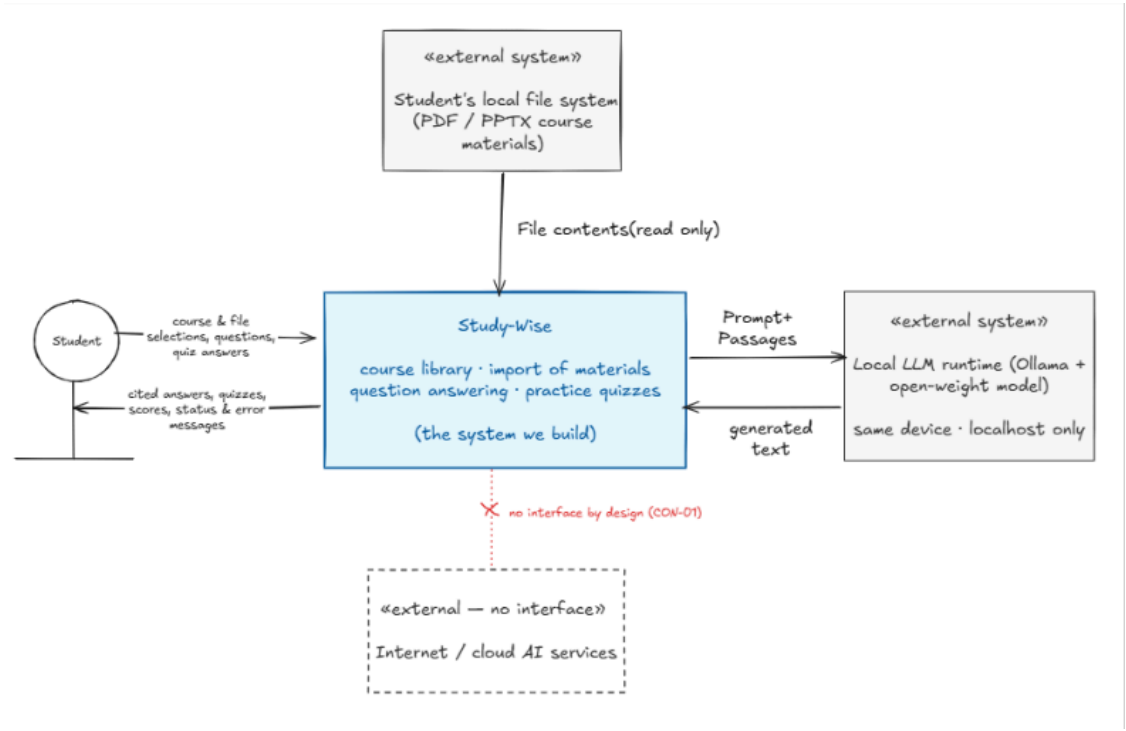
1.4 References

- **[R1]** ISO/IEC/IEEE 29148:2018, *Systems and software engineering — Life cycle processes — Requirements engineering*.
- **[R2]** CS 3354 Fall 2026 syllabus, Project Deliverable P1 template and P2 project description (Canvas).
- **[R3]** CS 3354 lecture material, Chapters 4–6: feasibility, requirements engineering, use cases.
- **[R4]** Ollama project documentation and README, github.com/ollama/ollama (MIT licence; stated memory guidance for running 7B-parameter models).
- **[R5]** The University of Texas at Dallas, Student Code of Conduct — academic integrity provisions (Office of Community Standards and Conduct).
- **[R6]** Licence terms of candidate open-weight models (e.g. Meta Llama community licence, Apache-2.0 for Mistral 7B).
- **[R7]** Study-Wise project README — product vision, in the team repository.
- **[R8]** CS 3354 lecture material, Chapters 7 and 8 (Lessons 13 to 16): use-case analysis, high-level design.
- **[R9]** Study-Wise benchmark set, /docs/benchmark in the team repository.

2 Overall Description

2.1 Product perspective and context

Unchanged since v0.6.



The boundary sits around the part that turns a student's files and questions into cited answers and quizzes. That is the part with no off-the-shelf equivalent that respects CON-01, so it is ours to build. The language model and the runtime that executes it are deliberately outside the box: we call them but do not build or train them, which is why they appear as an external system and as dependency A2. The student's file system is external because Study-Wise only reads files the student chooses and never modifies them. Instructors never interact with the system, but their material enters through the file-system interface, which is why they are stakeholders without being actors.

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
SH1: Students (primary users)	Import their materials, ask questions, take practice quizzes	Correct answers they can check against their own slides; when their materials do not cover a question, an answer from general knowledge clearly labelled as not from their materials (pending team decision on Q1); practice questions for exam prep; no cost. A need for privacy ("nothing uploaded") was not confirmed: all five interviewees already upload course files to cloud AI tools and none raised privacy.	Hardware and study survey (n = 10, 6–8 Nov 2026; one respondent is a team member) and student interviews (n = 5, 6–8 Nov 2026). Data and tallies in /docs/elicitation.
SH2: Instructors who author the materials (never use it)	Their slides and notes are the system's input	Their material is not reproduced, shared with anyone outside the class, or uploaded to any online environment	The CS 3354 instructor's reply to Q3: processing slides with a local tool for personal study is allowed under the course-materials clause; uploading them to a hosted service is not (CH-15).
SH3: UTD Office of Community Standards and Conduct (never uses it)	Enforces academic integrity	The tool supports practice and does not produce or submit graded work	Document review of the Student Code of Conduct [R5].

Stakeholder	Role in relation to the system	What they need from it	How you learned this
SH4: Model and runtime licensors (never use it)	Own the LLM runtime and model weights we depend on	Use within licence terms: attribution, acceptable-use rules, no redistribution of weights	Document review of licences [R4], [R6].
SH5: CS 3354 instructor and TAs	Set the deliverables and assess them	Deliverables on the syllabus dates; a repository they can open; requirements they can test	Syllabus, P1 template and P2 description [R2].
SH6: Development team (builders and maintainers)	Build, operate and test the system on their own laptops	A scope achievable by six students in one term with $0 budget	Team meeting 15 Oct 2026; feasibility in 2.6.

2.3 User characteristics

Technical students (CS, CE, SE and other STEM majors): Comfortable installing desktop software and following a README; all five interviewees already use cloud AI assistants (ChatGPT, Gemini, Claude) to study, and all five have received wrong answers from them. Some catch errors by checking their slides or noticing inconsistent follow-up answers; others find errors hard to spot. They therefore need to be told when an answer has no support in their materials, rather than receive a confident guess. Three of five interviewees said the slide or page an answer came from is very important and that they would open it to check. They use personal laptops, often on battery: 8 of 10 survey respondents study on battery at least some of the time. They use the system hardest in the days before an exam, when they are short on time and impatient with slow answers (QR-01).

Non-technical students (all other majors): Can import files and type questions but cannot be assumed to use a terminal, install a runtime, or diagnose why a model is not running. The survey shows this is a skill gap rather than a major: 4 of 10 respondents have never installed software from a README or terminal, and these include a Computer Science student. This user class is why FR-15 exists and why first-time setup is a use case of its own (UC-06, Q5).

Constraints from both classes: Survey respondents' laptops range from 8 GB or less (3 of 10) to 32 GB or more (3 of 10); 6 of 10 have 16 GB or more, so assumption A1 holds, but only narrowly. 9 of 10 run Windows 11 or macOS 14 or later (CON-03); only 3 of 10 confirmed a separate graphics card; none reported less than 10 GB of free storage. 2 of 10 respondents use assistive features while studying (zoom or large text, speech-to-text, a screen reader). We have not yet written an accessibility requirement and record this as open question Q10 rather than invent a threshold. The samples are small and mostly Computer Science students (7 of 10 survey respondents), so these results are indicative, not conclusive.

2.4 Constraints

Unchanged since v0.6.

- **C1 — No course material, question or generated content may leave the student's device.** Source: product vision [R7]; instructors' interest in their material (SH2). This removes every cloud-hosted model or storage option.
- **C2 — No paid services, subscriptions or API keys.** Source: product vision [R7]; students' cost need (SH1); team budget of $0 (SH6). This removes commercial model APIs even where they could be made private.
- **C3 — Must run on students' own laptops (Windows 11 or macOS 14+, no discrete GPU).** Source: students (SH1) — an assumption, see Q2. This removes models too large to run on that hardware.
- **C4 — Only components whose licences allow free use and publication of our source in a repository the instructor can open.** Source: licences [R4], [R6]; course repository requirement [R2].

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	Assumption: most target students have laptops with at least 16 GB RAM. Survey (CH-01): holds narrowly, 6 of 10 respondents have 16 GB or more and 3 of 10 have 8 GB or less.	S-01 showed what breaks: on an 8 GB laptop an 8B model does not run at all (the laptop froze), so 8 GB students need a 3B-class model with weaker answers (ADR-04, R1, R2).
A2	Dependency: a free local LLM runtime (Ollama [R4]) remains available for Windows and macOS and accepts requests from other programs on the same machine.	Q&A and quiz generation (FR-07, FR-10) have no model to call; we would have to switch runtimes, costing roughly a week of the schedule (contained by RuntimeGateway, ADR-02).
A3	Assumption: course materials are text-based PDFs and PPTX files, not scans.	Imports appear to succeed but yield no text, so questions find nothing; OCR (currently out of scope) would be needed. FR-03 detects this case.
A4	Confirmed (CH-15): processing instructor-provided materials locally for personal study is allowed by the course-materials clause, as long as nothing is uploaded to an online environment.	The product's main input would be disallowed and scope would shrink to student-written notes only. No longer expected to fail.
A5	Dependency: an open-weight model small enough for A1 hardware produces usable quiz questions and grounded answers.	QR-03 and QR-04 fail; a larger model would violate C3, so features would have to be cut back to Q&A only (R2).

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict
Technical	Feasible with a 3B-class model; 8B-class not feasible on 8 GB laptops; 16 GB not yet measured	Spike S-01 (section 8), 8 GB laptop: the 3B model answered 20 of 20 benchmark questions within 20 s (median 7.4 s, 18th-fastest 15.1 s); the 8B model took 486 s to load and then froze the laptop. Answer correctness, not speed, is now the open technical risk (R2).
Operational	Feasible for technical students; uncertain for non-technical	Survey (CH-01): 4 of 10 respondents have never installed software from a README or terminal, including a Computer Science student, so setup support is needed (FR-15, UC-06). The setup trials planned before P2 were not run and are planned before P3 (Q5, R4, CH-11).
Schedule	Feasible for the MUST set	11 MUST requirements; the highest-effort features (OCR, sync, Canvas integration) are excluded in 1.2. The plan in 10.2 has dated milestones with owners through P3.
Economic / resource	Feasible	$0 cost: the runtime is MIT-licensed [R4]; candidate models are free to use [R6]; GitHub is free for students; development hardware is team-owned laptops.

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one "shall" sentence	Priority	Source	Acceptance criterion
FR-01	The system shall import a PDF file selected by the student into a course the student chooses.	MUST	SH1, R7	Import a 30-page text PDF into course "CS 3354"; the file appears in that course's material list with 30 pages.
FR-02	The system shall import a PowerPoint (.pptx) file selected by the student into a course the student chooses.	MUST	SH1, R7	Import a 40-slide .pptx; the file appears in the course's material list with 40 slides.
FR-03	The system shall reject an imported file that contains no extractable text with a message naming that file.	SHOULD	A3	Import an image-only scanned PDF; a message names the file and says no text was found; the file is not listed.

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-04	The system shall let the student create a course with a name of their choice.	MUST	SH1	Create course “CS 4349”; it appears in the course list.
FR-05	The system shall keep courses and imported materials available after the application is closed and reopened.	MUST	SH1	Import a file, close and reopen the application; the file is listed and questions about it are answered without re-importing.
FR-06	The system shall remove a material the student selects so that it is no longer used in answers or quizzes.	SHOULD	SH1	Remove file X, then ask a question answered only in X; the system reports that no relevant material was found (FR-09).
FR-07	The system shall answer a typed question using only passages from the materials of the course the student selected.	MUST	SH1, R7	For each of the 20 benchmark questions, every cited source belongs to the selected course.
FR-08	The system shall display with every answer the file name and page or slide number of each passage used.	MUST	SH1	All 20 benchmark answers show at least one citation; opening each cited page shows the quoted passage.
FR-09	The system shall state that no relevant material was found, instead of answering, when no passage in the selected course matches the question.	MUST	SH1, SH3	Ask 5 off-topic questions (e.g. “Who won the 2022 World Cup?”) in a CS 3354 course; all 5 return the no-material message.
FR-10	The system shall generate a multiple-choice practice quiz of between 1 and 20 questions, as the student chooses, from a selected course.	MUST	SH1, R7	Request 10 questions; 10 questions appear, each with 4 options of which exactly one is marked correct.
FR-11	The system shall generate short-answer practice questions with a model answer when the student chooses that format.	COULD	R7	Request 5 short-answer questions; 5 questions appear, each with a model answer and a citation.
FR-12	The system shall limit quiz questions to the materials the student selects within a course.	SHOULD	SH1	Select only “Lecture 5.pdf”; every generated question cites Lecture 5.pdf.
FR-13	The system shall report the number of correct answers out of the total when the student submits a multiple-choice quiz.	MUST	SH1	Answer a 10-question quiz with 7 known-correct choices; the system reports 7 / 10.
FR-14	The system shall show for each submitted quiz question the correct answer with its source file and page or slide number.	MUST	SH1	Submit a quiz; every question shows the correct option plus a citation that opens to the supporting passage.
FR-15	The system shall tell the student, with steps to fix it, when the local LLM runtime is not running or has no model installed.	MUST	SH1, A2	Stop the runtime, then ask a question; a message says the runtime is not running and lists the steps to start it.
FR-16	The system shall reject a selected file that it cannot open, with a message naming that file. (New, CH-06)	SHOULD	UC-01 A3	Select a password-protected PDF; a message names the file; the file is not listed.
FR-17	The system shall keep each submitted quiz with its questions, the student’s choices and the score, so the student can review it later. (New, CH-20)	SHOULD	Q7	Submit a 5-question quiz, close and reopen the application; the quiz is listed with its score and each question shows the student’s choice and the correct option.
FR-18	The system shall, when the student imports a file whose name matches a material already in the course, ask whether to replace it and, on confirmation, replace the old material. (New, CH-21)	SHOULD	Q8	Import “Lecture 5.pdf” twice; the second import asks to replace; after confirming, the course lists one “Lecture 5.pdf” and answers cite the new file.

3.2 Quality requirements

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	Performance: 90% of answers shall be fully displayed within 20 seconds of submission on the reference laptop with a model of up to 8B parameters.	SHOULD	SH1, A1	Scripted run of the 20-question benchmark set; log submit-to-complete time; pass if the 18th-fastest time is ≤ 20 s. S-01 (section 8): met with the 3B model on an 8

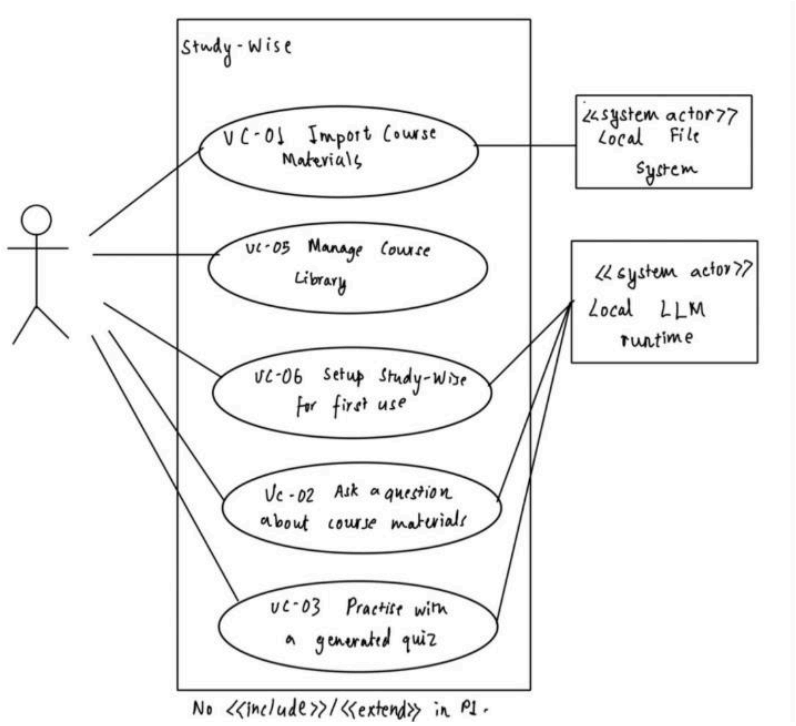
ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
				GB laptop (18th-fastest 15.1 s); not yet measured on the 16 GB reference laptop (CH-05). Design driver QAS-01.
QR-02	Performance: a 50-page text PDF shall become available for questions within 90 seconds of import on the reference laptop.	SHOULD	SH1	Import the same 50-page PDF 5 times on a freshly started application; pass if the median time from import to "ready" is ≤ 90 s. Design driver QAS-02.
QR-03	Answer quality: at least 85% of answers to the benchmark set shall be correct and supported by their cited passage.	MUST	SH1, SH3	Two team members score each answer independently against the answer key and cited page; a third resolves disagreements; pass if ≥ 17 / 20. S-01 saved the 3B answers for this scoring; one reviewer's provisional read is about 13 / 20 (R2).
QR-04	Quiz quality: at least 80% of generated multiple-choice questions shall have exactly one defensible correct option.	MUST	SH1	Generate 30 questions from 3 different courses' materials; two reviewers judge each independently; pass if ≥ 24 / 30 agreed valid.
QR-05	Usability: 4 of 5 first-time student testers shall import a file and receive a cited answer within 10 minutes without help.	SHOULD	SH1, 2.3	Moderated session with 5 students not on the team, starting after installation; time to first cited answer, with no hints given.

3.3 Constraint requirements

Unchanged since v0.6.

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The system shall not transmit course materials, questions, answers or quiz content to any destination outside the student's device.	C1; R7; SH2	Privacy is the reason the product exists; without it Study-Wise is a weaker copy of cloud tools. Checked by capturing network traffic (e.g. Wireshark) during a 30-minute scripted session: no packets from Study-Wise to non-loopback addresses (QAS-03).
CON-02	The system shall operate without any paid service, subscription or API key.	C2; SH1; SH6	Target users and the team have no budget; a paid dependency would block both use and development.
CON-03	The system shall run on student-owned laptops with Windows 11 or macOS 14 or later and no discrete GPU.	C3; SH1 (assumed, Q2)	Students will not buy hardware to study; a system that needs a GPU excludes most of them.
CON-04	The system shall use only third-party components whose licences permit free use and publication of the project source.	C4; R4; R6; R2	The course requires an instructor-accessible repository, and a licence violation cannot be fixed after release.

3.4 Initial use cases



Six candidate use cases were drawn; one (UC-04 Take Quiz) was merged into UC-03, leaving the five shown. Each one starts with the student and ends when the student has something of value: materials ready, a library organized, a working setup, an answer, or a scored practice quiz.

UC-01 Import Course Materials · UC-02 Ask a Question About Course Materials · UC-03 Practice with a Generated Quiz · UC-05 Manage Course Library (create courses, remove materials) · UC-06 Set Up Study-Wise for First Use.

Note on relationships and numbering. We first drew “Generate Quiz” (UC-03) and “Take Quiz” (UC-04) separately, linked by «include». A generated quiz that is never taken gives the student nothing, so generating a quiz is a step, not a goal. We merged them into UC-03 and retired the number UC-04 so identifiers stay stable. No other «include» or «extend» relationship survived the same test. “Search materials” and “Call model” are steps inside UC-02 and UC-03, not use cases.

UC-01, UC-02 and UC-03 are expanded in section 4 (CH-07).

3.5 Requirements to use case traceability

The P1 matrix is extended in section 9.1, and its findings are carried into section 9.2 (CH-07).

4 Use-Case Analysis

4.1 Use cases selected for analysis

Use case	Name	Why it was selected	Non-trivial steps
UC-02	Ask a Question About Course Materials	Carries most of the product’s value; touches the most quality requirements (QR-01, QR-03, CON-01); depends on an external system whose failure the student can see. Specified in full at P1.	(4)
UC-03	Practice with a Generated Quiz	Second core goal; QR-04 (quiz quality) is the requirement we are least sure a small local model can meet (R2). Specified in full at P1.	(6), (8), (10)

Use case	Name	Why it was selected	Non-trivial steps
UC-01	Import Course Materials	Every other use case depends on it; realizes five MUST/SHOULD requirements (FR-01 to FR-05) and QR-02; involves an external system (the student's file system) and a failure the student can see (FR-03).	(4), (6)

UC-06 Set Up Study-Wise for First Use was considered but not analyzed: how the model is obtained during setup is still open (Q5), and expanding it now would mean inventing the answer. See 9.2.

4.2 Expanded use cases

UC-02 Ask a Question About Course Materials

Field	Content
Primary actor	Student
Supporting actors and external systems	Local LLM runtime (external system)
TUCBW	The student chooses to ask a question in one of their courses.
TUCEW	The student sees an answer with the source of each passage it used, or a statement that no material in the course covers the question.
Preconditions	Study-Wise is installed on the student's device.
Postconditions	The student has seen either an answer citing at least one passage from the selected course, or an explicit statement that none was found. No question, passage or answer has left the device. The course's materials are unchanged.
Traces to requirements	FR-07, FR-08, FR-09, FR-15, QR-01, QR-03, CON-01

Actor input / action	System response
	(0) The system shows the student's courses, each with its name and number of materials.
(1) The student selects a course.	(2) The system shows the course name and its materials, each with its name and number of pages or slides, and invites a question about them.
(3) The student types a question and submits it.	(4) The system shows an answer built only from passages in the selected course, and, for each passage used, the material name and the page or slide number.
(5) The student reads the answer and the sources it cites. This is the acknowledgment.	

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	(0)	The student has no courses.	The system says there are no courses yet and offers to create one (UC-05).	Ends
A2	(2)	The selected course has no materials.	The system says the course has no materials and offers to import some (UC-01).	Ends
A3	(3)	The question is empty.	The system asks the student to type a question.	Rejoins (3)
A4	(4)	No passage in the course relates to the question.	The system states that no material in this course covers the question and gives no answer (FR-09).	Ends
A5	(4)	The local LLM runtime is not running or has no model.	The system says the runtime is unavailable and lists the steps to fix it (FR-15). The student fixes it and resubmits, or leaves.	Rejoins (3), or ends
A6	(4)	No answer after 60 seconds.	The system shows that it is still working and offers to cancel. On cancel, no answer is shown.	Rejoins (4), or ends

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A7	(5)	The student selects one of the cited sources.	The system shows that page or slide of the material, with the passage used marked.	Ends

UC-03 Practice with a Generated Quiz

Field	Content
Primary actor	Student
Supporting actors and external systems	Local LLM runtime (external system)
TUCBW	The student chooses to start a practice quiz in one of their courses.
TUCEW	The student sees their score and, for every question, the correct answer with its source.
Preconditions	Study-Wise is installed on the student's device.
Postconditions	The student has seen a score (correct out of total) and, for every question, the correct option with its material name and page or slide number. The quiz and its score are kept for later review (FR-17). No material, question or answer has left the device. The course's materials are unchanged.
Traces to requirements	FR-10, FR-11, FR-12, FR-13, FR-14, FR-15, FR-17, QR-04, CON-01

Actor input / action	System response
	(0) The system shows the student's courses, each with its name and number of materials.
(1) The student selects a course to practise in.	(2) The system shows the course's materials, all chosen by default.
(3) The student chooses which materials the quiz should draw from.	(4) The system shows the chosen materials and asks for a question format and a number of questions from 1 to 20.
(5) The student chooses multiple choice and a number of questions.	(6) The system shows that many questions, each with four options, drawn only from the chosen materials.
(7) For each question, the student chooses one option.	(8) The system marks the chosen option for that question.
(9) The student submits the quiz.	(10) The system shows the number of correct answers out of the total and, for each question, the correct option with its material name and page or slide number.
(11) The student reads their score and the correct answers. This is the acknowledgment.	

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	(0)	The student has no courses.	The system says there are no courses yet and offers to create one (UC-05).	Ends
A2	(2)	The selected course has no materials.	The system says the course has no materials and offers to import some (UC-01).	Ends
A3	(3)	The student chooses no materials.	The system asks for at least one material.	Rejoins (3)
A4	(5)	The number is outside 1 to 20.	The system shows the allowed range.	Rejoins (5)
A5	(5)	The student chooses short answer (FR-11).	The system shows the questions; the student writes answers and submits; the system shows a model answer with its source for each question, with no score (Q6).	Ends
A6	(6)	The local LLM runtime is not running or has no model.	As UC-02 A5 (FR-15).	Rejoins (5), or ends

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A7	(6)	The chosen materials are too short for the requested number.	The system shows as many questions as it could produce and states the actual number (assumption pending Q4).	Rejoins (7)
A8	(9)	Some questions are unanswered.	The system lists the unanswered questions. The student answers them, or submits anyway and they count as incorrect.	Rejoins (7), or (10)
A9	before (9)	The student abandons the quiz.	The system confirms the quiz was discarded; no score is shown.	Ends

UC-01 Import Course Materials

Field	Content
Primary actor	Student
Supporting actors and external systems	Student's local file system (external system)
TUCBW	The student chooses to add materials to a course.
TUCEW	The student sees each imported material listed in the course with its number of pages or slides and marked ready for questions.
Preconditions	Study-Wise is installed on the student's device.
Postconditions	Each imported material is listed in the course with its page or slide count and can be used by UC-02 and UC-03, including after Study-Wise is closed and reopened. The student's original files are unchanged. Nothing has left the device.
Traces to requirements	FR-01, FR-02, FR-03, FR-04, FR-05, FR-16, FR-18, QR-02, QR-05, CON-01

Actor input / action	System response
	(0) The system shows the student's courses, each with its name and number of materials.
(1) The student selects a course.	(2) The system shows the course's materials, each with its name and number of pages or slides.
(3) The student chooses to add materials.	(4) The system shows the PDF and PowerPoint files on the student's device.
(5) The student selects one or more files.	(6) The system shows each selected file in the course's materials with its number of pages or slides, and marks it ready for questions.
(7) The student sees the new materials listed as ready. This is the acknowledgment.	

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	(0) or (1)	The student has no course for these materials.	The system asks for a course name; the student enters one; the system shows the new, empty course (FR-04).	Rejoins (2)
A2	(5)	A selected file contains no text that can be extracted.	The system names the file, says no text was found, and does not list it (FR-03). Other selected files continue.	Rejoins (6)
A3	(5)	A selected file cannot be opened (damaged or password-protected).	The system names the file and says it could not be opened; the file is not listed. Other files continue. (No requirement yet; see 4.3.)	Rejoins (6)
A4	(5)	A selected file has the same name as a material already in the course.	The system says the file is already in the course and asks whether to replace it. On yes, the old material is replaced; on no, the file is skipped (FR-18; students' answer to Q8).	Rejoins (6)

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A5	(5)	The student cancels without selecting a file.	The system shows the course's materials unchanged.	Ends

4.3 Self-review log

Each expanded use case was run against the ten questions of the Lesson 13 review checklist by a member who did not write it. Checks with no defect are listed once at the end.

Use case	Checklist question	Defect found	How it was fixed
UC-02, UC-03	6. Are the pre-conditions things you may assume, not failures you must handle?	P1 pre-condition "the selected course contains at least one imported material" can be false in normal use, and P1 already handled it as flow 1a, so the two contradicted each other.	Removed from pre-conditions; kept as alternate flow A2.
UC-02, UC-03	6. (as above)	P1 pre-condition "Study-Wise is set up (UC-06)" can become false at any time if the runtime stops.	Removed; handled by UC-02 A5 and UC-03 A6 (FR-15).
All three	2. Is there a step (0)?	P1 use cases had no step (0).	Added step (0) to each: the courses with their material counts.
UC-02	3. Does every row hold exactly one turn?	P1 steps 3 to 5 (find passages, send to runtime, runtime returns) were three system steps with no actor turn between them.	Merged into one system response, (4); the internal steps are split out in the 5.2 scenario table.
UC-03	3. (as above)	P1 step 6, "answers every question and submits the quiz," merged two actor actions.	Split into (7)/(8) and (9)/(10).
UC-02	5. Does it end with an acknowledgment the actor can perceive?	P1 ended with "reads the answer and, optionally, opens a cited source," which mixed the acknowledgment with an optional action.	Main flow ends at (5), the student reading the cited answer; opening a source became A7.
UC-01	8. Is every alternate named by the step it branches from? Does each trace to a requirement?	A3 (file cannot be opened) has no requirement behind it.	Proposed FR-16 through change control (0.2).
UC-01	8. (as above)	A4 (file already in course): the right behavior is a choice for users, not for us.	Recorded as open question Q8 instead of inventing an answer. Q8 was later answered by interviews; A4 now follows FR-18 (CH-21).
UC-03	7. Is there a post-condition a tester could turn into a test?	Whether a finished quiz and its score are kept is not stated anywhere.	Post-condition limited to what the student has seen; retention recorded as open question Q7. Q7 was later answered by interviews, and the postcondition now includes FR-17 (CH-20).
All three	1, 9, 10	None found. The actor is the role Student; nothing happens that the student neither sees nor waits for (importing in UC-01 is waited for, per QR-02); external systems are named by role, not product, and no screen widgets or stores are named.	—
All three	4. Does every item the actor supplies exist as a domain model attribute?	Checked against the domain model in 5.3 (K1).	See 5.3.

5 Analysis Models

5.1 Domain model

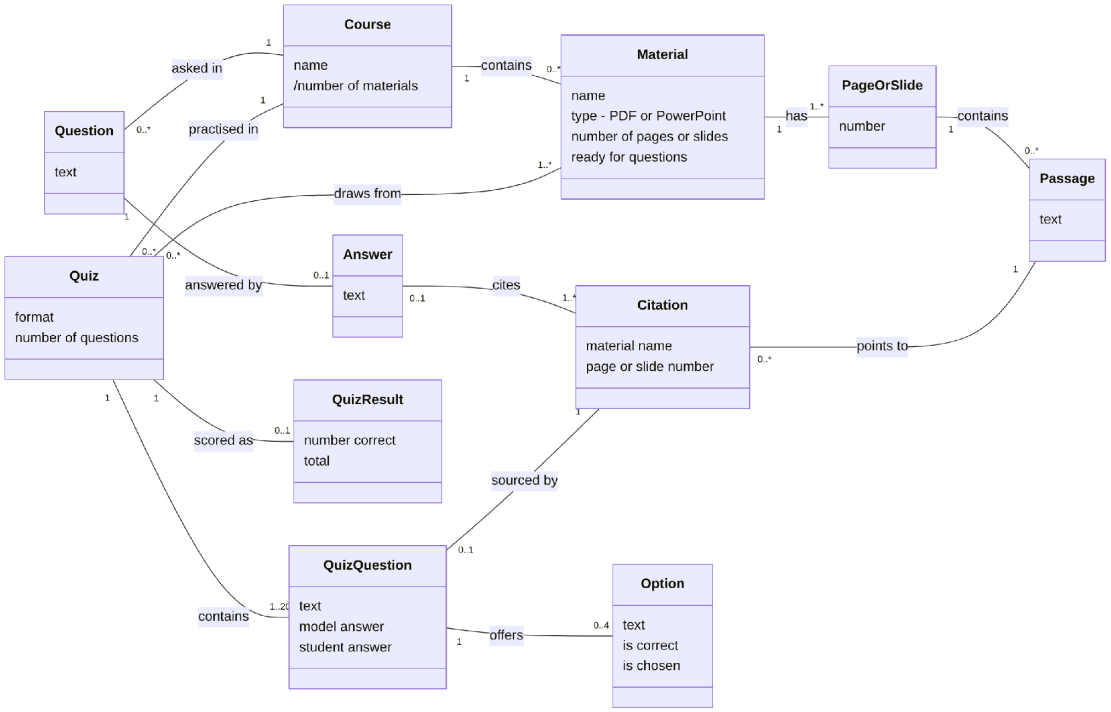


Figure 5.1 Domain model (conceptual classes, attributes, associations with multiplicities). Analysis model: no operations, no data types, no storage. Source: `diagrams/domain-model.mmd`.

Class	What it represents	Key attributes	Associations (with multiplicity)
Course	A named collection of materials the student creates (section 1.3)	name, /number of materials (derived)	contains 0..* Material; 0..* Question asked in it; 0..* Quiz practised in it
Material	One PDF or PowerPoint file imported into a course	name, type (PDF or PowerPoint), number of pages or slides, ready for questions	belongs to 1 Course; has 1..* PageOrSlide; drawn from by 0..* Quiz
PageOrSlide	One page of a PDF or one slide of a PowerPoint	number	belongs to 1 Material; contains 0..* Passage
Passage	A piece of text from one page or slide, the unit an answer or question is built from	text	lies on 1 PageOrSlide; pointed to by 0..* Citation
Question	A question the student types about one course	text	asked in 1 Course; answered by 0..1 Answer (none when no material covers it)
Answer	The response shown to the student	text	answers 1 Question; cites 1..* Citation
Citation	Where a fact came from	material name, page or slide number	points to 1 Passage; belongs to 0..1 Answer or 0..1 QuizQuestion
Quiz	One practice quiz the student starts	format (multiple choice or short answer), number of questions	practised in 1 Course; draws from 1..* Material; contains 1..20 QuizQuestion; scored as 0..1 QuizResult
QuizQuestion	One question in a quiz	text, model answer (short answer only), student answer (short answer only)	in 1 Quiz; offers 0..4 Option (4 for multiple choice, 0 for short answer); sourced by 1 Citation

Class	What it represents	Key attributes	Associations (with multiplicity)
Option	One choice in a multiple-choice question	text, is correct, is chosen	belongs to 1 QuizQuestion
QuizResult	The score of a submitted multiple-choice quiz	number correct, total	scores 1 Quiz

"Chosen materials" in UC-03 is modelled as the association Quiz *draws from* Material, not as an attribute, because it names other objects rather than a value (see K1 in 5.3).

5.2 Object interaction models

Three analysis objects are added to the domain classes. They are not data; each one controls or supports an interaction:

Interaction object	Why it exists (one line)
QuestionHandler	In control of UC-02: turns a question into a cited answer or a "no material" statement.
QuizMaster	In control of UC-03: produces a quiz from chosen materials, records the student's choices and scores it.
PassageFinder	Finds the passages in a course, or in chosen materials, most relevant to a question or topic. Used by both use cases.

The local LLM runtime appears on both diagrams as an **external system** (outside our boundary, section 2.1). It is named by role, not by product.

5.2.1 UC-02 Ask a Question About Course Materials

(a) Non-trivial steps. Only step (4) needs objects inside the system: passages must be found, relevance decided, an answer generated and its citations recorded. The others are trivial: (0) and (2) display courses and materials the system already holds; (1) and (3) are actor input that becomes message 3; (5) is the student reading, with nothing computed.

(b) Scenario table. Each row was first written as a sentence ("The QuestionHandler asks the PassageFinder for the passages in the course most relevant to the question").

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
3	Student	submits a question	question text, selected course	QuestionHandler
4a	QuestionHandler	asks for the passages in the course most relevant to the question	question text, course	PassageFinder
4b	QuestionHandler	checks that at least one passage is relevant	the passages found	QuestionHandler (itself)
4c	QuestionHandler	asks for an answer built only from those passages	question text, passages	Local LLM runtime
4d	QuestionHandler	records the answer with a citation for each passage used	generated text, passages	Answer
4e	QuestionHandler	shows the answer with its citations	Answer, Citations	Student

Row 4b is where alternate flow A4 (no relevant passage, FR-09) branches off: if the check fails, 4c to 4e do not happen and the "no material" statement is shown instead. Row 4b is a self-call, a sign that QuestionHandler is deciding alone; Chapter 8 keeps that rule inside AnswerService (6.3).

(c) Analysis sequence diagram.

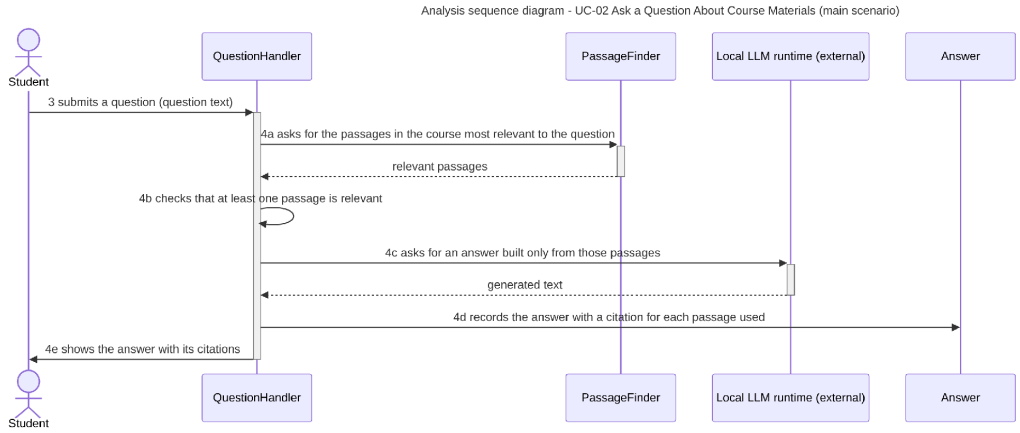


Figure 5.2.1 Analysis sequence diagram for UC-02 (analysis). Message numbers match the scenario table; returns are dashed and unnumbered. Source: `diagrams/seq-uc02.mmd`.

5.2.2 UC-03 Practice with a Generated Quiz

(a) Non-trivial steps. (6): questions must be drawn from passages, generated, checked and recorded. (8): the chosen option must be recorded, which changes data. (10): the score must be computed and recorded. The others are trivial: (0), (2) and (4) display what the system already holds; (1), (3), (5), (7) and (9) are actor input, with (5), (7) and (9) becoming messages; (11) is the student reading.

(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
5	Student	chooses multiple choice and a number of questions	format, number of questions, chosen materials	QuizMaster
6a	QuizMaster	checks the number is between 1 and 20	number of questions	QuizMaster (itself)
6b	QuizMaster	asks for passages from the chosen materials	chosen materials, number of questions	PassageFinder
6c	QuizMaster	asks for that many questions, each with four options and one correct, from the passages	passages, number of questions	Local LLM runtime
6d	QuizMaster	checks each question has exactly one correct option and a source passage	generated questions, passages	QuizMaster (itself)
6e	QuizMaster	records the questions with their options and citations	checked questions	Quiz
6f	QuizMaster	shows the questions with their options	Quiz, QuizQuestions, Options	Student
7	Student	chooses an option for a question	the chosen option	QuizMaster
8	QuizMaster	records the option as chosen	the chosen option	Option
9	Student	submits the quiz	the quiz	QuizMaster
10a	QuizMaster	asks how many chosen options are correct	QuizQuestions, Options	Quiz
10b	QuizMaster	records the number correct and the total	number correct, number of questions	QuizResult
10c	QuizMaster	shows the score and each correct option with its source	QuizResult, correct Options, Citations	Student

Row 6d exists because the runtime is outside our trust boundary (6.4): what it returns is checked before the student sees it (ADR-05, QR-04). Steps (7)/(8) repeat once per question; the table shows one pass.

(c) Analysis sequence diagram.

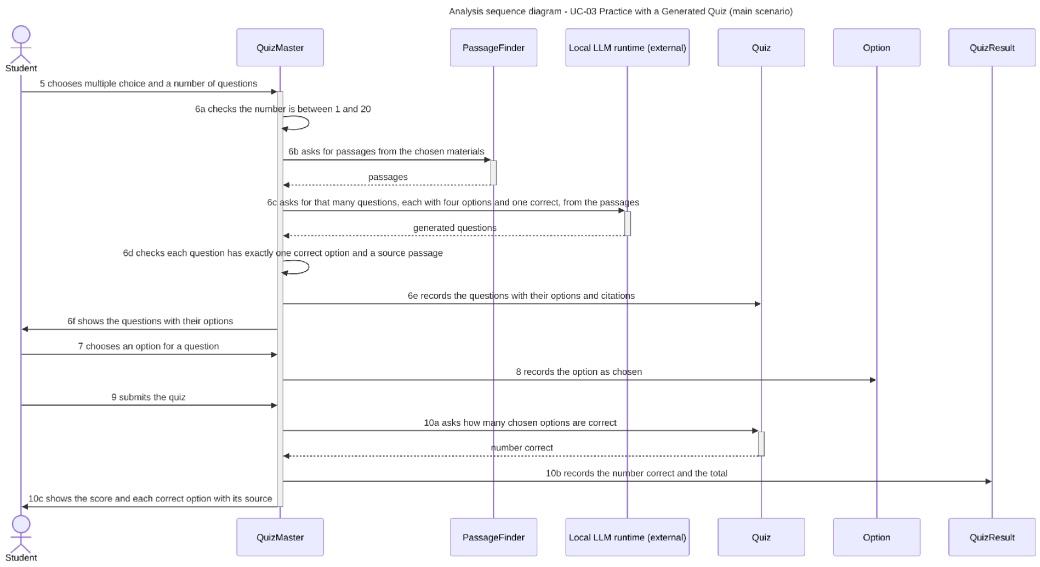


Figure 5.2.2 Analysis sequence diagram for UC-03 (analysis). Message numbers match the scenario table. Source: `diagrams/seq-uc03.mmd`.

5.3 Consistency cross-check

Check	Result (pass, or what failed)	Fixed in
K1 Every item in an actor column (4.2) is an attribute in the domain model (5.1)	**Failed, fixed.** UC-03 A5 has the student write short answers, but no class held what the student wrote. Also, “chosen materials” (UC-03 step 3) is a set of materials, not a value.	Added *student answer* to QuizQuestion; chosen materials modelled as the association Quiz *draws from* Material (note under 5.1).
K2 Every object on a sequence diagram is a domain class or a declared interaction object	Pass. QuestionHandler, QuizMaster and PassageFinder are declared in 5.2; Answer, Quiz, Option and QuizResult are in 5.1; the local LLM runtime is the external system from 2.1.	—
K3 Every message on a sequence diagram corresponds to exactly one scenario-table row	Pass. 6 messages for 6 rows (UC-02), 14 for 14 (UC-03), same numbers; returns unnumbered.	—
K4 Every non-trivial step in 5.2 (a) has at least one scenario-table row	**Failed, fixed.** Section 4.1 listed UC-03’s non-trivial steps as (6) and (10). Step (8), recording the chosen option, changes data, so it is non-trivial (Lesson 14, slide 19) and needed a row.	4.1 corrected to (6), (8), (10); row 8 added.
K5 Every analyzed use case traces to at least one functional requirement	Pass. UC-02: FR-07, FR-08, FR-09, FR-15. UC-03: FR-10 to FR-15. UC-01 (analyzed in 4.2, not carried into 5.2): FR-01 to FR-05, FR-16.	—
K6 The last messages of each diagram make the use case postcondition true	Pass for the main scenarios: UC-02 4d/4e give a shown answer citing at least one passage; UC-03 10b/10c give a shown score and the correct option with its source for every question. The other half of UC-02’s postcondition (“or an explicit statement that none was found”) is reached through 4b and A4, not drawn. The “nothing left the device” part cannot be shown at analysis level; it is carried by the deployment view (6.4) and QAS-03.	—

6 High-Level Design (Architecture)

6.1 Architectural drivers: quality-attribute scenarios

QAS	Quality attribute	Stimulus	Artifact	Response	Response measure	From
QAS-01	Performance	A student submits one of the 20 benchmark questions on the reference laptop, with a model of up to 8B parameters and no other heavy application open	AnswerService, PassageIndex, RuntimeGateway and the local runtime	The answer is fully displayed with its citations	At least 18 of 20 answers within 20 s. S-01: met with the 3B model on an 8 GB laptop (20 of 20; 18th-fastest 15.1 s)	QR-01
QAS-02	Performance	A student imports a 50-page text PDF into a course on a freshly started application	MaterialIngestor, PassageIndex, LocalStore	The material is listed as ready for questions	Median of 5 imports at most 90 s	QR-02
QAS-03	Security (privacy)	A scripted 30-minute session: import two decks, ask the 20 benchmark questions, take one 10-question quiz, while network traffic is captured	The whole system as deployed (6.4)	No course material, question, answer or quiz content is sent off the device	0 packets from Study-Wise processes to any non-loopback address	CON-01
QAS-04	Availability	The local runtime is stopped while Study-Wise is running, and the student asks a question	RuntimeGateway, AnswerService, StudyUI	The student is told the runtime is not running, with the steps to start it; the application keeps running	Message shown within 5 s; 0 materials lost; the same question succeeds once the runtime is restarted	FR-15 (measure added, CH-09)

Constraints that shape the architecture. CON-01 puts every component and every stored byte on the student's device and allows only loopback communication. CON-03 (laptop, no GPU) limits the model to roughly 8B parameters or fewer and makes generation the slowest step. CON-02 and CON-04 limit us to free, openly licensed components (Ollama, MIT; SQLite, public domain; pypdf, BSD; python-pptx, MIT).

6.2 Architectural style and alternatives

Style considered	Drivers it serves	What it costs	Verdict
Local client–server with a layered server: a browser user interface talks to one Python backend process on the same laptop over loopback; the backend is layered (API, application, domain, data and integration)	QAS-03: everything stays on the device. Modifiability: a layer changes behind its interface; the runtime and store sit behind single components (ADR-02, ADR-03). QAS-01: one backend process, so passages and prompts never cross a process boundary except to the runtime.	Two programs plus the runtime must be running (setup burden, R4). The loopback port can be reached by other programs on the laptop (R7). The layering is relaxed: some arrows skip a layer (see below).	**Chosen** (ADR-01)
Cloud client–server: hosted backend and model, thin client	Easier setup, faster models on server GPUs, one place to upgrade	Sends course material off the device, breaking CON-01 and QAS-03; hosting and model APIs cost money, breaking CON-02	Rejected
Microservices: ingestion, retrieval, answering and quizzes as separate processes	Independent deployment and scaling	A student would run four or more processes; every boundary on the answer path adds a cross-process hop and serialization of passages (against QAS-01); far more setup and failure modes for one laptop and six students	Rejected

Style considered	Drivers it serves	What it costs	Verdict
Single-process desktop application (one program with its own window)	Simplest install, no port exposed	Fixes the user interface toolkit to the backend language; the team's README and skills already assume a web front end and a Python backend; harder to keep presentation separate from rules	Rejected

Why this style. The deciding driver is QAS-03: anything that leaves the laptop is out. Students did not ask for privacy (2.2); the driver comes from the instructors (SH2), whose course-materials clause forbids uploading slides to online environments (CH-15). That rule removes the cloud style before performance is considered. Among the on-device styles, QAS-01 favors keeping retrieval, prompting and citation inside one process, which rules out microservices. Client–server keeps the user interface separate from the rules (separation of concerns) at the cost of one loopback hop that is small next to generation time (S-01 measures the whole path). Inside the backend, **pipe-and-filter** is used for one job only, importing: extract text, split into passages, compute vectors, store. **The layering is relaxed and we say so:** a layer may call any layer below it, never above. Three arrows skip a layer (LocalApi to CourseLibrary, the two services to RuntimeGateway); routing them through an intermediate layer would add components with no responsibility of their own (KISS). This is recorded as a cost in ADR-01.

6.3 Component view

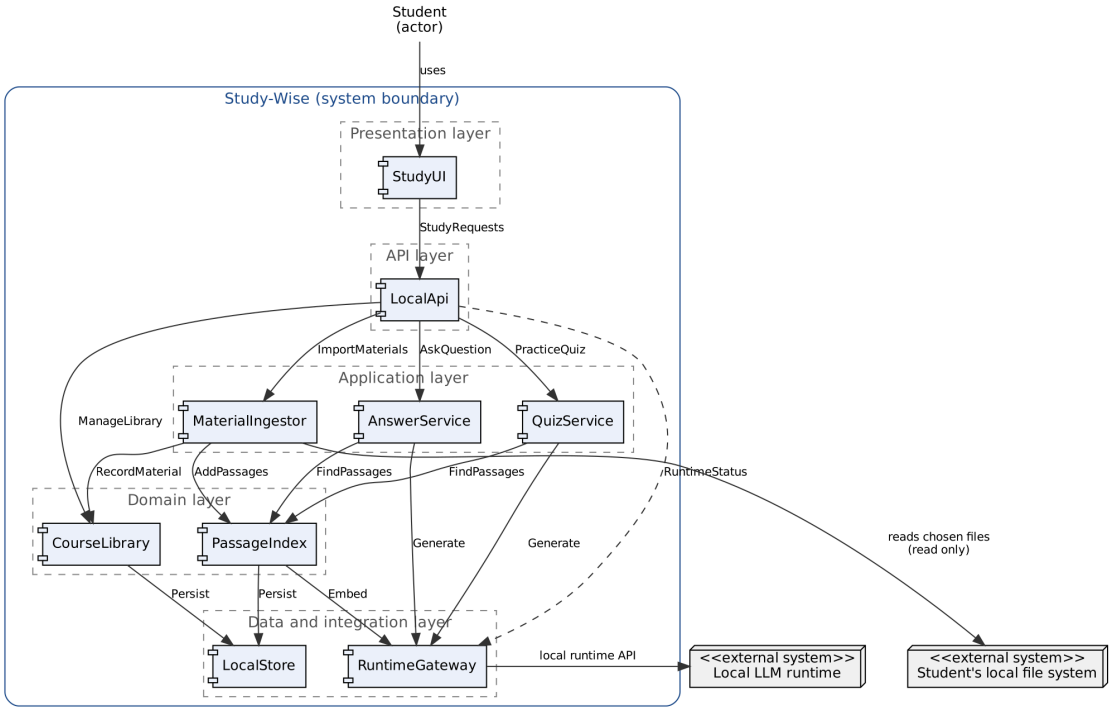


Figure 6.3 Component diagram: components, required interfaces (arrow labels), external systems outside the boundary. Source: `diagrams/component-view.dot`.

Component	Responsibility (one sentence)	Provides	Requires	Answers to (UC / QAS / ADR)
StudyUI	Shows the student their courses, answers and quizzes and passes their actions to the backend.	screens to the Student	StudyRequests	UC-01, UC-02, UC-03; QAS-04 (shows the runtime message); ADR-01
LocalApi	Accepts the user interface's requests on the loopback interface and routes each to the service that owns it.	StudyRequests	ImportMaterials, AskQuestion, PracticeQuiz,	ADR-01; QAS-03 (binds to 127.0.0.1 only)

Component	Responsibility (one sentence)	Provides	Requires	Answers to (UC / QAS / ADR)
			ManageLibrary, RuntimeStatus	
MaterialIngestor	Turns a chosen PDF or PowerPoint file into numbered pages or slides and passages.	ImportMaterials	RecordMaterial, AddPassages, file read	UC-01; FR-01 to FR-03, FR-16; QAS-02
AnswerService	Answers a question only from the selected course's passages, with citations, or says none were found. (Realizes QuestionHandler.)	AskQuestion	FindPassages, Generate	UC-02; QAS-01; ADR-04
QuizService	Generates, checks and scores practice quizzes from chosen materials. (Realizes QuizMaster.)	PracticeQuiz	FindPassages, Generate	UC-03; ADR-05
CourseLibrary	Keeps the student's courses and the materials in each.	ManageLibrary, RecordMaterial	Persist	UC-01, UC-05; FR-04 to FR-06
PassageIndex	Holds every passage with its vector and returns the most relevant passages for a question within a course or chosen materials. (Realizes PassageFinder.)	AddPassages, FindPassages	Embed, Persist	UC-02, UC-03; QAS-01, QAS-02; ADR-03
LocalStore	Saves and reloads the library and passages on the device.	Persist	(local file)	FR-05; ADR-03
Runtime Gateway	Is the only component that talks to the local LLM runtime, and reports whether it is available.	Generate, Embed, RuntimeStatus	local runtime API	ADR-02; QAS-04; FR-15

External systems (outside the boundary): the local LLM runtime (called only through RuntimeGateway) and the student's file system (read only, by MaterialIngestor).

Coupling test (Lesson 15). Changing how PassageIndex ranks passages, how LocalStore lays out its file, or which runtime RuntimeGateway calls requires opening 0 other components, because each is reached only through a question-style interface ("find passages for this question in this course"), not by fetching its data.

6.4 Deployment view

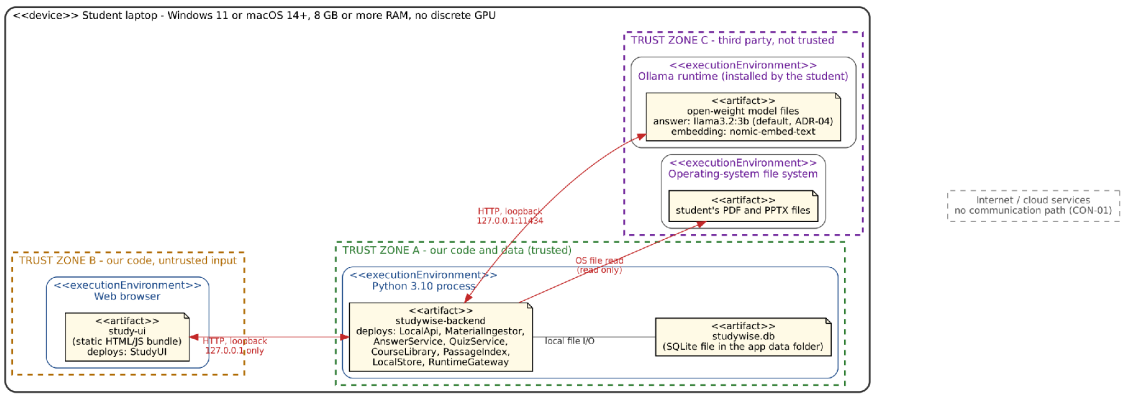


Figure 6.4 UML deployment diagram: one device node, its execution environments and artifacts, communication paths with protocols, and trust zones. Source: `diagrams/deployment-view.dot`.

Node / execution environment	What runs on it (artifacts, components)	Communicates with (protocol)	Trust zone
«device» Student laptop	Everything below. No server, no cloud.	Nothing off the device (CON-01, QAS-03)	—

Node / execution environment	What runs on it (artifacts, components)	Communicates with (protocol)	Trust zone
«executionEnvironment» Web browser	study-ui (static HTML/JS bundle, served by the backend): StudyUI	Backend: HTTP over loopback, 127.0.0.1 only	B: our code, but its input is not trusted (any program on the laptop can send requests to the loopback port)
«executionEnvironment» Python 3.10 process	studywise-backend: LocalApi, MaterialIngestor, AnswerService, QuizService, CourseLibrary, PassageIndex, LocalStore, RuntimeGateway; studywise.db (SQLite file in the application data folder)	Browser: HTTP, loopback. Runtime: HTTP to 127.0.0.1:11434. File system: operating-system file read.	A: trusted
«executionEnvironment» Ollama runtime (installed separately by the student)	Answer model (3B-class by default, ADR-04) and embedding model. On Apple-silicon laptops it uses the integrated GPU and shared memory, which is why model size and RAM are linked (S-01).	Backend: HTTP, loopback	C: third party, not trusted
«executionEnvironment» Operating-system file system	The student's PDF and PowerPoint files	Backend: read only	C: not trusted (files may be malformed or protected)

Trust boundaries and what crossing each one costs. - B to A (browser to backend): LocalApi validates every request and binds to 127.0.0.1, never to all interfaces, so nothing on the campus network can reach it (R7). - C to A (runtime to backend): generated text is checked before it is shown; quiz questions must have exactly one correct option and a real source passage (row 6d, ADR-05). - C to A (files to backend): a file that cannot be opened or has no text is rejected with a message (FR-03, FR-16), never trusted to be well formed.

What happens when each path is cut. Backend to runtime: QAS-04, the student sees the runtime message and nothing is lost. Browser to backend: the user interface cannot do anything and shows that the backend is not running (setup issue, Q5). Backend to file system: only an import fails, with FR-16's message.

6.5 From analysis to architecture

Analysis object (5.1 and 5.2)	Component that owns it (6.3)	Note
Course, Material, PageOrSlide	CourseLibrary	Stored through LocalStore
Passage	PassageIndex	Created by MaterialIngestor, then handed to PassageIndex
PassageFinder	PassageIndex	Interaction object realized by the component
Citation	PassageIndex	Built from a passage's material name and page number; carried inside Answer and QuizQuestion
Question, Answer	AnswerService	Not stored after display (no requirement asks for history)
QuestionHandler	AnswerService	Interaction object realized by the component
Quiz, QuizQuestion, Option, QuizResult	QuizService	Kept after submission through LocalStore (FR-17, CH-20)
QuizMaster	QuizService	Interaction object realized by the component
Local LLM runtime	External	Reached only through RuntimeGateway
Student (actor)	StudyUI	The actor interacts through StudyUI

6.6 Design principles applied

Principle	Where it shows in our design	Evidence
Separation of concerns	Presentation (StudyUI), request routing (LocalApi), rules (services) and storage (LocalStore) are separate components in separate layers	6.3, ADR-01

Principle	Where it shows in our design	Evidence
Information hiding	RuntimeGateway hides the runtime's API and the model name; PassageIndex hides how passages are ranked and how vectors are stored; LocalStore hides the file layout	Generate, FindPassages, Persist interfaces; ADR-02, ADR-03
Low coupling	Services ask PassageIndex a question ("most relevant passages for this question in this course") and receive only passages; changing ranking opens 0 other components	6.3 coupling test
High cohesion	Every responsibility in 6.3 is one sentence about one subject; importing, answering and quizzing are separate components rather than one "AI service"	6.3 table
Design for change	The most likely changes (runtime, model, retrieval method) each sit behind one component; a model change is a setting, not code	ADR-02, ADR-03, ADR-04
Keep it simple (KISS)	One backend process, one database file, no separate vector database, no message bus; relaxed layering instead of pass-through components	ADR-01, ADR-03

7 Design Rationale

7.1 Architecture decision records

ADR-01 Local client–server with a relaxed layered backend

Field	Content
Status	Accepted 8 Nov 2026
Date and owner	Drafted 6 Nov 2026 by Lokesh Reddy Penugonda; accepted by the team 8 Nov 2026
Context (drivers)	QAS-03 and CON-01 forbid anything leaving the device. QAS-01 makes the answer path time-critical on a laptop with no GPU (CON-03). The README already assumes a web front end and a Python backend.
Options considered	(a) Cloud client–server with a hosted model. (b) Local client–server: browser UI and one Python backend on loopback, backend layered. (c) Microservices: one local process per component. (d) Single-process desktop application.
Decision	We will build (b): a browser user interface served by one Python backend process bound to 127.0.0.1, with the backend in four layers (API, application, domain, data and integration). A layer may call any layer below it, never above.
Consequences	Favors privacy (QAS-03) and modifiability (presentation separate from rules). Costs: two programs plus the runtime must be running, which adds to setup (R4, Q5); the loopback port is reachable by other programs on the same laptop, so every request is validated (R7); layering is relaxed, so three arrows skip a layer and a change to RuntimeGateway's interface can reach the application layer directly. (a) was rejected because it breaks CON-01 and CON-02; (c) adds processes and cross-process hops on the answer path; (d) ties the interface to the backend language and mixes presentation with rules.
Evidence	README [R7]; S-01 measures the full answer path on this structure.
Related	QAS-01, QAS-03, R4, R7

ADR-02 Reach the local LLM runtime only through RuntimeGateway, using Ollama's local HTTP API

Field	Content
Status	Accepted 8 Nov 2026
Date and owner	Drafted 6 Nov 2026 by Lokesh Reddy Penugonda; accepted by the team 8 Nov 2026
Context (drivers)	Two services need generated text and PassageIndex needs vectors. FR-15 and QAS-04 need one place that knows whether the runtime is running. The runtime is third-party software outside our trust boundary (dependency A2).
Options considered	(a) Each service calls Ollama directly. (b) All calls go through one RuntimeGateway component that calls Ollama over HTTP on 127.0.0.1:11434. (c) Load the model inside the backend process with an embedded inference library, with no separate runtime.
Decision	We will use (b).

Field	Content
Consequences	Favors modifiability: switching runtime or model touches one component (design for change), and FR-15's check lives in one place. Costs: one loopback HTTP hop per call (expected to be milliseconds next to seconds of generation; S-01 measures the total), and the student must install and start Ollama separately (A2, Q5, R4). (a) spreads the runtime's API across three components; (c) removes a separate install but makes our process manage gigabytes of model memory and threads, and makes the backend crash when the model runs out of memory.
Evidence	Ollama documentation [R4] (MIT licence, Windows and macOS builds). S-01 called Ollama over loopback for every answer and still met QAS-01 with the 3B model, so the gateway hop is not the bottleneck.
Related	QAS-01, QAS-04, ADR-04, R1

ADR-03 Keep the library and passage vectors in one local SQLite file, with similarity search inside the backend

Field	Content
Status	Accepted 8 Nov 2026
Date and owner	Drafted 6 Nov 2026 by Lokesh Reddy Penugonda; accepted by the team 8 Nov 2026
Context (drivers)	FR-05 needs courses and materials to survive restarts. FR-07 and FR-09 need the most relevant passages within one course. QAS-01 limits retrieval time. CON-02 and CON-04 require free, openly licensed parts. A course is small: two lecture decks give 71 pages (S-01 dry run).
Options considered	(a) A dedicated vector database library (for example Chroma or FAISS). (b) One SQLite file holding courses, materials, passages and their vectors, with similarity computed in the backend over the selected course's passages only. (c) Plain JSON files.
Decision	We will use (b).
Consequences	Favors simplicity (KISS): one dependency already in Python, one file to back up or delete, transactions so a crash mid-import does not corrupt the library. Costs: search time grows linearly with the passages in a course; past a few thousand passages per course it may become noticeable. Fitness check for P3: retrieval for a 20-deck course stays under 1 s, otherwise revisit (a). (a) is faster at scale but adds an install-heavy dependency for courses that are small; (c) has no transactions, so an interrupted import can break FR-05.
Evidence	S-01: 71 pages embedded and indexed in 5.5 s; one vector per page, cosine similarity computed in process, top 3 pages retrieved. Expected page retrieved for 17 of 20 questions; smaller chunks are a P3 option.
Related	QAS-01, QAS-02, R8

ADR-04 Default answer model: a 3B-class open-weight model

Field	Content
Status	Accepted 8 Nov 2026 for the default; the 8B option on 16 GB laptops is revisited after the S-01 rerun (10.2)
Date and owner	Drafted 7 Nov 2026 by Lokesh Reddy Penugonda from S-01 results; accepted by the team 8 Nov 2026
Context (drivers)	QAS-01 (answer within 20 s) pulls toward a small model; QR-03 and QR-04 (answer and quiz quality) pull toward a larger one. CON-03: no discrete GPU, 8 to 16 GB RAM. Risk R1.
Options considered	(a) An 8B-class instruction model (llama3.1:8b, 4-bit). (b) A 3B-class model (llama3.2:3b, 4-bit). (c) Let the student choose any model.
Decision	Default to (b). Offer (a) only on laptops with at least 16 GB, once a second S-01 run on a 16 GB laptop shows it meets QAS-01. The model name is a setting read by RuntimeGateway.
Consequences	Favors speed and the 8 GB students: the 3B model met QAS-01 with room to spare. Costs answer quality: one reviewer's provisional read of the 3B answers is about 13 of 20 correct, below QR-03's 17, so R2 is now the main technical risk, and retrieval fixes (smaller chunks, more pages, a reranker) come before any larger model is reconsidered. (a) does not run on 8 GB at all. (c) was rejected as the default because untested models make QR-03 and QR-04 unverifiable.
Evidence	S-01 (section 8): 3B, 20 of 20 under 20 s, median 7.4 s; 8B, 486 s to load, then the 8 GB laptop froze.
Related	QAS-01, ADR-02, R1, R2, CH-14, Q9

ADR-05 Check everything the runtime returns before the student sees it

Field	Content
Status	Accepted 8 Nov 2026
Date and owner	Drafted 6 Nov 2026 by Lokesh Reddy Penugonda; accepted by the team 8 Nov 2026
Context (drivers)	The runtime is in trust zone C (6.4). A small model can return a quiz question with two correct options, no correct option, or a source that does not exist, which would break QR-04 and FR-14.
Options considered	(a) Show the runtime's output as returned. (b) Ask the model to check its own output. (c) Check in QuizService and AnswerService: every quiz question must have four options, exactly one correct, and a citation to a passage that was sent; every answer citation must name a passage that was sent; failing questions are regenerated once, then dropped and the student told the actual number.
Decision	We will use (c).
Consequences	Favors correctness (QR-04, FR-08, FR-14). Costs: extra generation time whenever a question is regenerated, which pushes against QAS-01-style response times for quizzes, and the student may receive fewer questions than asked (the same behavior as UC-03 A7, pending Q4). (a) trusts a third party; (b) costs a second model call per question and is checked by the same model that made the mistake.
Evidence	None yet; QR-04's review at P3 will show how often questions are dropped.
Related	QR-04, UC-03 row 6d, R6

ADR-06 Refuse off-topic questions with a similarity gate followed by a relevance check

Field	Content
Status	Accepted 8 Nov 2026
Date and owner	Drafted 7 Nov 2026 by Lokesh Reddy Penugonda from S-01 results; accepted by the team 8 Nov 2026
Context (drivers)	FR-09: the system must say no material was found rather than answer an off-topic question. In S-01, "Explain how a TCP three-way handshake works" scored 0.632 against the decks, higher than the real benchmark question B04 (0.605), and was answered from general knowledge. No single similarity threshold separates the two.
Options considered	(a) Similarity threshold only (S-01's 0.55 gate). (b) Gate first, then ask the model a separate yes/no question: do these passages answer the question? Answer only on yes. (c) A reranker model that scores each passage against the question.
Decision	We will use (b): AnswerService keeps the cheap gate to refuse clearly unrelated questions without calling the model, and adds the yes/no check (scenario row 4b) before generating.
Consequences	Favors FR-09 and academic integrity (SH3). Costs one extra short model call for every question that passes the gate, which uses part of QAS-01's 20 s budget (S-01's 18th-fastest answer left about 5 s). (a) is measured to fail; (c) adds another model to download and run on 8 GB laptops.
Evidence	S-01: off-topic questions refused 4 of 5 with the gate and prompt alone. The yes/no check is not yet measured. Q1 (CH-19): if the team adopts labelled general-knowledge answers, the same check decides whether to label an answer instead of refusing it, so the decision holds either way.
Related	FR-09, QAS-01, CH-17

7.2 Trade-off summary

Quality favored	Architectural move	Cost accepted	Recorded in
Privacy (QAS-03)	Everything on the student's laptop, loopback only	Students run the backend and the runtime themselves; no help from a server's GPU	ADR-01
Modifiability	Relaxed layers; runtime behind RuntimeGateway	Some arrows skip a layer; one extra loopback hop per model call	ADR-01, ADR-02
Simplicity	One SQLite file, similarity computed in process	Linear search time as a course grows	ADR-03
Answer speed (QAS-01)	3B-class default model, chosen from S-01 timings	Weaker answers: QR-03 at risk (R2)	ADR-04

Quality favored	Architectural move	Cost accepted	Recorded in
Correctness (QR-04)	Validate the runtime's output at the trust boundary	Extra time when questions are regenerated; possibly fewer questions	ADR-05
Refusing off-topic questions (FR-09)	Similarity gate plus a yes/no relevance check	One more model call per answered question	ADR-06

8 Prototype Evidence: Architecture Spike

Field	Content
Spike ID and title	S-01 Can a student laptop run the model fast enough?
Risk or decision it addresses	R1 (laptops too slow), QAS-01 (from QR-01), ADR-04 (model choice); also FR-09 (off-topic refusal) and the answer path of ADR-02 and ADR-03
Question (answerable yes or no)	On a laptop with no discrete GPU, does a local model, given the 3 most relevant lecture pages, answer at least 18 of the 20 benchmark questions within 20 seconds? Asked for an 8B and a 3B model.
Set-up	Python harness `run_benchmark.py` (PyMuPDF and the standard library). Ollama 0.34.3 on 127.0.0.1; embedding model nomic-embed-text; chat models llama3.2:3b and llama3.1:8b, both 4-bit. Data: the benchmark set (`docs/benchmark`, PR #30) and its two lecture decks, 71 pages (decks not committed, SH2). One chunk per page, cosine similarity, top 3 pages passed to the model; a 0.55 similarity gate refuses without calling the model. Machine: MacBook Air M3, 8 GB unified memory, macOS 26.0.1, the 8 GB laptop R1 asks for (not the 16 GB reference laptop). Simulated: no user interface and no store.
How it was measured	Time from submitting each question to the last token of the answer, including query embedding and retrieval; model loading (cold start) reported separately. Also recorded per question: best similarity score, pages retrieved, whether the expected page was retrieved, and the full answer for QR-03 scoring. The 5 off-topic questions from the benchmark README were run for FR-09.
Result (numbers)	**llama3.2:3b:** 20 of 20 within 20 s; median 7.4 s; 18th-fastest 15.1 s; slowest 18.1 s; cold start 13.2 s; 71 pages indexed in 5.5 s; expected page retrieved for 17 of 20; off-topic questions refused 4 of 5. **llama3.1:8b:** 486 s to load and give a first answer, then the laptop froze from swapping and had to be restarted; no benchmark answer completed. **Answer: yes for 3B, no for 8B on 8 GB.** Correctness, one reviewer's provisional read: about 13 of 20 (not the QR-03 result, which needs two reviewers).
What it changed or confirmed	Confirmed QR-01 and QAS-01 for a 3B model on 8 GB. Changed ADR-04 to a 3B default (CH-14). Raised R2: correctness, not speed, is now the main risk. Led to ADR-06 (FR-09 needs more than a threshold, CH-17) and risk R9 (broken ligatures in one deck's text, CH-18). Still open: the same run on a 16 GB laptop with both models.
Where to see it	`/spikes/S-01/` (moved from `/spike-01`): `run_benchmark.py`, `README.md`, and `results/` (per-question CSV and summary JSON for each run); PR #32, commit bf4e95e

9 Traceability

9.1 Extended traceability matrix

Replaces the P1 matrix (old section 3.5). Use cases in bold were analyzed in sections 4 and 5. Analysis objects in italics are interaction objects (5.2).

Requirement	Use case	Analysis classes	Component(s)	QAS / ADR	Evidence
FR-01	**UC-01**	Material, PageOrSlide, Passage	MaterialIngestor, CourseLibrary, PassageIndex	QAS-02, ADR-03	—
FR-02	**UC-01**	Material, PageOrSlide, Passage	MaterialIngestor, CourseLibrary, PassageIndex	QAS-02, ADR-03	—
FR-03	**UC-01** A2	Material	MaterialIngestor	—	—

Requirement	Use case	Analysis classes	Component(s)	QAS / ADR	Evidence
FR-04	**UC-01** A1; UC-05	Course	CourseLibrary	—	—
FR-05	**UC-01** postcondition; UC-05	Course, Material, Passage	LocalStore, CourseLibrary	ADR-03	—
FR-06	UC-05	Material, Passage	CourseLibrary, PassageIndex	ADR-03	—
FR-07	**UC-02**	Question, Passage, Answer, *QuestionHandler*, *PassageFinder*	AnswerService, PassageIndex	QAS-01, ADR-02, ADR-04	Benchmark (PR #30); S-01
FR-08	**UC-02**	Answer, Citation	AnswerService, PassageIndex	ADR-05	Benchmark (PR #30)
FR-09	**UC-02** A4 (row 4b)	*QuestionHandler*	AnswerService	ADR-06	S-01: off-topic refused 4 of 5
FR-10	**UC-03**	Quiz, QuizQuestion, Option, *QuizMaster*, *PassageFinder*	QuizService, PassageIndex	ADR-05	—
FR-11	**UC-03** A5	QuizQuestion (model answer, student answer)	QuizService	—	—
FR-12	**UC-03** (3), row 6b	Quiz *draws from* Material, *PassageFinder*	QuizService, PassageIndex	—	—
FR-13	**UC-03** (10), rows 10a–10b	QuizResult, Option	QuizService	—	—
FR-14	**UC-03** (10), row 10c	Option, Citation	QuizService	ADR-05	—
FR-15	**UC-02** A5; **UC-03** A6; UC-06	Local LLM runtime (external)	RuntimeGateway, StudyUI	QAS-04, ADR-02	—
FR-16	**UC-01** A3	Material	MaterialIngestor	—	—
FR-17	**UC-03** postcondition	Quiz, QuizResult	QuizService, LocalStore	—	Interviews, Q7
FR-18	**UC-01** A4	Material	MaterialIngestor, CourseLibrary	—	Interviews, Q8
QR-01	**UC-02**	—	AnswerService, PassageIndex, RuntimeGateway	QAS-01, ADR-02, ADR-04	S-01: 20 of 20 within 20 s (3B, 8 GB)
QR-02	**UC-01**	—	MaterialIngestor, PassageIndex, LocalStore	QAS-02, ADR-03	Partial: S-01 indexed 71 pages in 5.5 s (no import flow or store)
QR-03	**UC-02**	Answer, Citation	AnswerService	ADR-04, ADR-05, ADR-06	Benchmark (PR #30); S-01 answers saved; provisional read about 13 of 20
QR-04	**UC-03**	QuizQuestion, Option	QuizService	ADR-04, ADR-05	Not yet measured
QR-05	**UC-01**, **UC-02**	—	StudyUI	—	Not yet measured (usability sessions, P3)
CON-01	All	—	All; LocalApi binds to loopback	QAS-03, ADR-01	Deployment view (6.4)
CON-02	All	—	All	ADR-02, ADR-03	Licences named in 6.1

Requirement	Use case	Analysis classes	Component(s)	QAS / ADR	Evidence
CON-03	All	—	RuntimeGateway (model setting)	ADR-04	S-01: 8B does not run on 8 GB; 3B does
CON-04	None (build constraint)	—	All	ADR-02, ADR-03	Licences named in 6.1; licence list file not yet in the repository

9.2 Gap analysis

Gap	Items	What we did about it
Requirements with no component	None	Every FR, QR and CON reaches at least one component.
Components with no requirement, QAS or ADR	None. LocalApi answers only to ADR-01 and QAS-03, not to a functional requirement.	Accepted: it exists because of the client–server decision, and ADR-01 records why.
Use cases not analyzed (and why that is acceptable)	UC-05 Manage Course Library; UC-06 Set Up Study-Wise	UC-05 is two simple MUST/SHOULD operations (FR-04, FR-06) with no external system; its requirements still reach CourseLibrary. UC-06 depends on Q5 (does Study-Wise download the model, or does the student?); expanding it now would invent the answer. Both are expanded at P3.
Requirements with no evidence yet, or failing evidence	QR-02 (partial), QR-03 (provisional 13 of 20), QR-04, QR-05, CON-04	QR-03: two-reviewer scoring of the S-01 answers, then retrieval fixes (smaller chunks, more pages, a reranker). QR-02 and QR-04 are measured once MaterialIngestor and QuizService exist (10.2). QR-05 needs the usability sessions at P3. CON-04: a licence list is added to the repository at P3, and it must settle PyMuPDF (AGPL-3.0, used in S-01) before it enters the product.
Flow with no requirement (carried from P1)	UC-03 A7: too little material for the requested number of questions	Still waiting on Q4; ADR-05 uses the same behavior (generate fewer and say so) as the interim assumption.
Flows that became requirements during analysis	UC-01 A3 (file cannot be opened); UC-01 A4 and the UC-03 postcondition, once Q8 and Q7 were answered	Became FR-16 (CH-06), FR-18 (CH-21) and FR-17 (CH-20).

Findings carried from the P1 matrix, still valid: FR-15 exists because UC-06 had no requirement behind it; FR-05 is traced through postconditions because no actor starts it; “Generate Quiz” and “Take Quiz” were merged into UC-03.

10 Project Plan (updated; was P1 section 4)

10.1 Roles and working agreement

Unchanged since v0.6, with one correction: the P1 risk table called Malik Coleman “Terence Coleman” (CH-12). How the P2 drafting was shared is recorded in 0.1 and 10.5.

10.2 Plan to P3

Milestone	Date	Owner	Done when
S-01 rerun on a 16 GB laptop with both models; Q1 to Q9 answers folded into sections 1 to 4 through change control	13 Nov	Lokesh Reddy Penugonda	Results committed to `/spikes/S-01/results`; each answered question has a CH entry
QR-03 two-reviewer scoring of the S-01 answers; retrieval fixes tried	13 Nov	Nandhan Alahari, Malik Coleman	Both reviewers’ scores committed; result recorded in 3.2
Detailed design (Chapter 9): design classes and design sequence diagrams for UC-02 and UC-03	16 Nov	Nandhan Alahari, Shivansh Batra	Merged to `/docs/p3` through reviewed PRs

Milestone	Date	Owner	Done when
MaterialIngestor, CourseLibrary and LocalStore working; QAS-02 measured	19 Nov	Pranab Neupane	A 50-page PDF imports; median of 5 timings committed
PassageIndex, AnswerService and RuntimeGateway working; QAS-01 and QAS-04 measured; QR-03 benchmark scored	22 Nov	Nandhan Alahari	Benchmark run committed with two members' scores
QuizService with ADR-05 checks; QR-04 review of 30 questions	25 Nov	Malik Coleman	Two reviewers' judgements committed
StudyUI connected to all three analyzed use cases	25 Nov	Shivansh Batra	UC-01, UC-02 and UC-03 run end to end on one laptop
Test plan (Chapter 11); QAS-03 traffic capture	29 Nov	Malik Coleman	Test plan merged; capture shows 0 non-loopback packets
Two setup trials and five usability sessions (QR-05); UC-05 and UC-06 expanded	30 Nov	Lavanya Bansal	Notes committed to `/docs/elicitation`; Q5 answered
P3 submitted	9 Dec 2026	Shivansh Batra	PDF on Canvas matches the tagged commit

10.3 Risks (updated)

Risk	Status since P1	Likelihood / impact	Mitigation and owner
R1 Student laptops cannot run a usable model fast enough	Partly retired by S-01 (6 Nov): the 3B model meets QAS-01 on 8 GB; the 8B model does not run on 8 GB. 16 GB not yet measured.	Low for 3B / High	ADR-04: 3B default. Rerun S-01 on a 16 GB laptop by 13 Nov. Lokesh Reddy Penugonda.
R2 A small model gives wrong answers or quiz keys	Raised by S-01: a provisional read of the 3B answers is about 13 of 20, below QR-03's 17; one off-topic question was answered.	High / High	Two-reviewer QR-03 scoring; retrieval fixes (smaller chunks, more pages in the prompt, a reranker); ADR-05 checks quiz output; ADR-06 for off-topic questions. Nandhan Alahari (answers), Malik Coleman (quizzes).
R3 Instructor or UTD policy disallows processing course materials	Closed (CH-15): the instructor confirmed local processing for personal study is allowed; uploading to a hosted service is not, which CON-01 already forbids.	—	None needed.
R4 Non-technical students cannot complete setup	Open. The setup trials planned for 5 Nov were not run; moved to 30 Nov. ADR-01 and ADR-02 add to setup (a backend and a separately installed runtime).	Medium / Medium	Setup trials before P3; UC-06 written from them (Q5). Lavanya Bansal.
R5 A member becomes unavailable or misses section deadlines	Open, unchanged; name corrected (CH-12).	Medium / Medium	Issue board reviewed Mondays and Thursdays; any section with no commit for three days is reassigned, and Malik Coleman checks the reassignment appears in the repository history. Shivansh Batra.
R6 (new) The runtime returns malformed answers or quiz questions	New, from the trust boundary in 6.4.	Medium / Medium	ADR-05: QuizService and AnswerService check everything the runtime returns. Malik Coleman.
R7 (new) Another program on the laptop, or the campus network, reaches the backend's port	New, from ADR-01.	Low / High	LocalApi binds to 127.0.0.1 only and validates every request; checked in the QAS-03 capture. Shivansh Batra.
R8 (new) Similarity search becomes slow for very large courses	New, from ADR-03.	Low / Low	Fitness check at P3: retrieval for a 20-deck course under 1 s; otherwise revisit ADR-03. Pranab Neupane.

Risk	Status since P1	Likelihood / impact	Mitigation and owner
R9 (new) Lecture PDFs extract with broken text	New, from S-01: in the Chapter 8 deck, ligatures such as “fi” and “ti” come out as other characters (“files” becomes “Dles”), likely causing a retrieval miss. PyMuPDF and pypdf give the same result.	Medium / Medium	Normalize known ligature characters after extraction; test another extractor; add a garbled-text check to FR-03’s import test at P3. Pranab Neupane.

10.4 Repository

URL: https://github.com/shivbatra999/Study-Wise (public; Dr. Seker and the TAs can open it without an account).

Layout: - P1 section files at the repository root (`00-`… to `06-`…), P1 diagrams in `/docs/p1/diagrams` - `/docs/benchmark` — the 20-question benchmark set - `/docs/elicitation` — survey results and interview notes - `/docs/p2` — P2 section files and this PDF - `/docs/p2/diagrams` — diagram sources (`.mmd`, `.dot`) and the exported images - `/spikes/S-01` — spike script, README and results

Branches and pull requests: `main` holds reviewed work only. Each member works on a branch named `<initials>-<topic>` (for example `lp-uc-analysis`) and merges through a pull request reviewed by another member. Nobody commits directly to main. Commits are made from each member’s own GitHub account.

Deadline rule (CH-04): every section merged by Sunday 8 Nov, 6 PM. This PDF is committed to `/docs/p2`, the tag `p2` is placed on that commit and pushed (`git push origin p2`), and the same file is uploaded to Canvas by Lokesh Reddy Penugonda only.

10.5 Use of AI tools in P2

Member	Tool	What it was used for	What you checked or changed
Lokesh Reddy Penugonda	Claude (Anthropic)	Drafted the benchmark questions from two lecture decks; drafted first versions of sections 0 to 12 (except Lavanya’s elicitation text in 2.2, 2.3 and 11) and the diagram sources; applied the Lesson 13 checklist and the K1–K6 cross-checks to the drafts; drafted a spike script that was not used (S-01 is Nandhan’s harness). Sections were then handed to the owners listed in 0.1.	Checked every benchmark answer and page against the slides; cross-referenced every ID (FR, QR, UC, QAS, ADR, CH, R, Q) across sections 0 to 12; checked each sequence-diagram message against its scenario-table row and each component against 9.1; checked the numbers in sections 2 and 6 to 9 against the S-01 results and the elicitation files.
Lokesh Reddy Penugonda	Cursor	Ran the git branch, commit, pull-request, merge and tag steps for PRs #30 and #31, the P2 sections PR and the p2 tag.	Checked each diff contained only the intended files; removed the tool’s automatic co-author line.
Nandhan Alahari	Cursor agent, Claude Code (Anthropic)	Wrote and ran the S-01 harness and drafted the spike README; ran the Lesson 13 checklist on PR #31 and drafted the review comment (also logged in `data/ai-use-log.md`)	Timings come from the harness’s own logs, not the agent; benchmark answers and pages checked against the decks; each review finding checked against the PR; the correctness table is marked as one reviewer’s provisional read
Lavanya Bansal	Claude (Anthropic)	Drafted the elicitation summary and the updates to 2.2, 2.3 and section 11 from the survey and interview data (co-author trailer on commits 8a5f0de, 3cfda02, 22e588a, b3535c3)	Checked manually if data and pushed files are correctly merged

11 Open Questions (updated; was P1 section 5)

ID	Question	Raised by	Owner and due date	Status or answer	Assumption meanwhile
Q1	Should answers use only the student's materials, or also the model's general knowledge?	P1	Students via interviews, Lavanya Bansal, 7 Nov; requirement decision by the team, 13 Nov	**Answered 8 Nov:** 5 of 5 interviewees preferred a general-knowledge answer clearly labelled as not from their materials (1 said that under exam pressure they might prefer "not covered"). Requirement change pending a team decision (CH-19).	Materials only (FR-07, FR-09)
Q2	What RAM, operating system and GPU do target students have?	P1	Students via survey, Lavanya Bansal, 7 Nov	**Answered 8 Nov:** survey n = 10 (2.3). 6 of 10 have 16 GB or more, 3 of 10 have 8 GB or less; 9 of 10 run Windows 11 or macOS 14+; 3 of 10 confirmed a separate graphics card. A1 holds narrowly.	—
Q3	May students process instructor slides with a local AI tool for personal study?	P1	Dr. Seker; sent by Shivansh Batra 6 Nov	**Closed (CH-15).** Yes, with a local tool for personal study; uploading the decks to a hosted service is not allowed.	—
Q4	When chosen materials are too short for the requested number of quiz questions: fewer, repeat, or refuse?	P1	Students via interviews, Lavanya Bansal, 7 Nov	**Answered 8 Nov:** 3 of 5 chose fewer questions and a note; 2 of 5 chose the full number with repeated topics. The current behavior is confirmed; the FR will be written at P3 with the quiz design.	Generate fewer and say so (UC-03 A7, ADR-05)
Q5	Should Study-Wise download the model at setup, or does the student install the runtime and model?	P1	Non-technical students via setup trials, Lavanya Bansal, 30 Nov (moved from 5 Nov, CH-11)	Open	Student installs the runtime (ADR-02)
Q6	Should short answers be marked, and against what standard?	P1	The team (returned to us by the instructor as our design decision, CH-16); Lokesh Reddy Penugonda; decide by 16 Nov	Open. Students: 3 of 5 want a model answer with its source; 2 of 5 want marking but would only partly trust it. Options: (a) not marked, model answer and citation shown; (b) the model marks against the model answer, falling back to "not marked" below a confidence threshold; (c) the student marks their own answer against the model answer. Decision criterion: choose (b) only if it agrees with a human marker on at least 16 of 20 sample answers in a small spike; otherwise (a).	(a) Not marked (UC-03 A5)
Q7	Should a finished quiz and its score be kept for later review?	Analysis of UC-03 (4.3)	Students via interviews, Lavanya Bansal, 7 Nov	**Closed 8 Nov:** 3 of 3 interviewees asked want them kept. Became FR-17 (CH-20).	—
Q8	When a file already in the course is imported again: skip, replace, or add a second copy?	Analysis of UC-01 (4.3)	Students via interviews, Lavanya Bansal, 7 Nov	**Closed 8 Nov:** 3 of 3 interviewees asked said replace, because updated slides often keep the same name. Became FR-18 (CH-21).	—
Q9	Would students accept weaker answers from a	ADR-04	Students via interviews, Lavanya Bansal, 13 Nov	Open. S-01 showed the choice is forced on 8 GB laptops (8B does not run); the question now matters for 16 GB laptops.	3B default

ID	Question	Raised by	Owner and due date	Status or answer	Assumption meanwhile
	smaller model in exchange for speed?				(ADR-04)
Q10	Which accessibility features must Study-Wise support?	Survey (2.3)	Lavanya Bansal, before P3	Open: 2 of 10 respondents use assistive features (zoom or large text, speech-to-text, a screen reader).	Keyboard navigation only

12 Baseline v1.0 Declaration (was P1 section 6)

We declare this document, version 1.0, the baseline for P2. Sections 1 to 9 describe what we will build in P3. Any later change to them is recorded in section 0.2 with its reason.

Team member	Signature or initials	Date
Lavanya Bansal	Lavanya Bansal	November 8th
Lokesh Reddy Penugonda	Lokesh Reddy Penugonda	November 8th
Nandhan Alahari	Nandhan Alahari	November 8th
Pranab Neupane	Pranab Neupane	November 8th
Shivansh Batra	Shivansh Batra	November 8th
Malik Coleman	Malik Coleman	November 8th